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# Integrating agroforestry and perennial pastures to mitigate water logging and secondary salinity

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## Abstract

The studies at the 'Ucarro' site have identified rapid transmission of vertical drainage from the root zone to re-charge a transmissive aquifer that conveys virtually all the hill-slope outflow to the riparian zone of the local stream. This feature of a predominance of deep drainage in total outflow contrasts with other regional and national studies on duplex soils where lateral flow within the root zone constitutes a significant component of the total outflow. Hydraulic properties of the 'Ucarro' subsoil matrix are consistent with those of the other studies, suggesting that bypass flow, operative during phases of soil saturation, is the critical mechanism in determining this distinctive hydrologic response. These observations at 'Ucarro' serve to indicate that partitioning of rainfall excess between lateral and vertical flow is fraught with difficulty. Simulation of hydrologic response, reliant on hydraulic characterization of duplex soils for drainage prediction, needs to be qualified with large bounds of uncertainty.

Differences in annual evapotranspiration were evaluated for three categories of plant community to reveal a ranking according to depth of root uptake of soil water. The woody community used water in excess of annual rainfall by 100 mm due to uptake of groundwater. The herbaceous perennial, lucerne, used marginally less than annual rainfall whilst the annual pasture exploited less of the annual rainfall in most years. Additional evapotranspiration was achieved mainly during the summer/autumn season when differences in soil water uptake between communities were apparent. These seasonal differences in soil water uptake also explained the paradox that annual evapotranspiration was inversely related to above-ground growth and also to the maximum leaf area attained by the community during the year. Persistence of plant activity over the year was the major determinant of annual evapotranspiration and depth of rooting was a significant factor to extend the period of plant activity in supplying additional evapotranspiration. The amount of re-charge to groundwater for each community was approximately commensurate with the difference between annual values of rainfall and evapotranspiration with average annual values estimated to range from –100 to 35 mm. Calculations based on observed evapotranspiration by each category of community provided a range of hill-slope outflow for various combinations of tree cover, lucerne and annual agricultural communities as crop and pasture. An optimal mix comprising 12% tree

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cover, 30% lucerne and 58% annual crop and pasture offered a viable agricultural enterprise with outflow control at 5 mm per year, approaching that of the natural water balance. Integration of persistent perennial species, woody and herbaceous, with traditional agriculture provides satisfactory drainage control in this environment to ameliorate existing outbreaks of salinity. © 2002 Elsevier Science B.V. All rights reserved.

*Keywords:* Re-charge; Evapotranspiration; Water balance; Outflow; Drainage

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## 1. Introduction

Outbreaks of water logging, soil salinity and acidity are expressions of groundwater rise caused by accelerated vertical drainage from the root zone under current land management practices dedicated to food and fiber production. The shallow rooting habit associated with annual agricultural ecosystems is now accepted as a critical factor in the change in water balance that has given rise to land degradation following land clearing for agriculture (Hatton and Nulsen, 1999). The ability to retain excess rainfall for subsequent plant water use is much less under agriculture with shallow rooting annuals than under pristine ecosystems dominated by perennial species and supported by deep root systems that ensured continuing uptake and plant function across seasons and ensured the persistence of the ecosystems. The challenge to agriculture is to emulate the water control of natural ecosystems while maintaining the productivity of current agricultural systems. The papers in this publication describe studies undertaken at the 'Ucarro' farm, the site of innovative management designed to restrict the amount of water flowing to deep drainage and maintain farm prosperity. This paper emphasizes water control and the impact that changed control might have on farm productivity.

Improved water management at the 'Ucarro' farm has been achieved through a combination of engineering and biological interventions. The engineering solution is based on interceptor drains serving to mitigate water logging through enhanced transmission of shallow lateral flow and reduced vertical flow to deeper groundwater. Biological solutions involve the introduction of perennial species, woody and herbaceous, and are aimed at reducing deep drainage and groundwater accession through deeper rooting. Continuing uptake of soil water well into the summer season ensures persistence of the perennial species. The ability of these measures to modify groundwater response, either singly or in combination, has been evaluated through sets of detailed measurements that have determined the fate of rainfall on hill-slopes. An understanding has been gained on the impact of the interventions on the allocation of rainfall to groundwater or loss as water vapor into the atmosphere. The role of interventions in changing the proportion of total outflow between different forms of outflow, vertical and lateral, has also been addressed. As a starting point, the pathways of outflow and associated quantities of water that have been observed at the site will be summarized.

Evapotranspiration is the process with the major influence on the water balance as it disposes more than 80% of the annual rainfall. As the stomata exchange water vapor and carbon dioxide, growth responses should also be reflected in changes to the water balance. Maximizing leaf area is viewed as a key method of reducing drainage to

groundwater. Yet when measurements of annual evapotranspiration for tree strips, crops, subterranean clover and lucerne are compared, the relationship between net primary productivity and annual water use was not clear, while maximum leaf area and water use appeared to be inversely related. A method of characterizing daily evapotranspiration is advanced to explain differences in the annual values between communities and also to account for the counter-intuitive relationships with growth-related responses. Finally, noting that differences in evapotranspiration exist between plant communities, ways to incorporate attributes for water control by perennial species, herbaceous and woody, into an economically viable enterprise are explored. Using data of water use by the range of plant communities studied, hill-slope outflow from groundwater can be calculated for the various combinations of cover deployed in crop rotations with bordering contour strips of trees. These calculations are critical to the central issue of sustainable agriculture, namely the extent to which integrated productivity of herbage and crop yield is compromised in restoring drainage rates towards those experienced under natural vegetation.

## 2. Pathways of outflow

“Render, therefore, unto Caesar the things that are Caesar’s and to God the things that are God’s” (Matthew 22 v21).

The price of a healthy edaphic environment is a leaching fraction to remove those toxic salts that over time will impair plant growth. Stopping all leaching is counter-productive with salt accumulation. Excessive drainage causes the groundwater levels to rise and move salt, naturally occurring below the root zone, to within the zone of rooting. Clearly, a critical balance exists between vertical water flow in the soil and plant water uptake. Under the original dry sclerophyll forest of the region, the leaching fraction in the dissected landscape (McFarlane and Williamson, 2002) was likely to be in the range of 1–5 mm per year.

The observed outflow at the ‘Ucarro’ site is dominated by vertical drainage (Ward et al., 2001), in contrast to two other Australian case studies of water discharge from duplex soils similar to that at ‘Ucarro’ with an ‘A’ horizon of ~35 cm soil depth overlying massive clay (Table 1). A measured saturated hydraulic conductivity of less than 1 mm per day for the clay horizon in each of the three sites indicates limited transmission of soil

Table 1

Comparison of published information on hydrologic response of duplex soils at three instrumented sites across southern Australia, each with perennial pasture (Aston and Dunin, 1980; White et al., 2000b; Ward et al., 2001)<sup>a</sup>

| Site                      | Pasture  | Slope (%) | Scale (ha) | Years | $P_{av}$ (mm) | $Q_{av}$ (mm) | $Q/P$ (%) | $Q_D$ (%) | $Q_L$ (%) |
|---------------------------|----------|-----------|------------|-------|---------------|---------------|-----------|-----------|-----------|
| Krawarree (35.8°S149.6°E) | Themeda  | 3–10      | 5          | 71–76 | 964           | 265           | 27        | Trace     | 100       |
| BookBook (35.4°S147.5°E)  | Phalaris | 2–5       | 0.1        | 94–97 | 546           | 77            | 14        | 43        | 57        |
| Ucarro (33.8°S117.5°E)    | Lucerne  | 2–8       | 4          | 95–98 | 404           | 10            | 2         | 90        | 10        |

<sup>a</sup> Annual rainfall ( $P_{av}$ ) and total outflow ( $Q_{av}$ ) are shown as averaged values for the specified years of observation together with the proportional contribution ( $Q_D$ ) from vertical deep drainage and combined lateral flows of surface runoff and interflow ( $Q_L$ ).

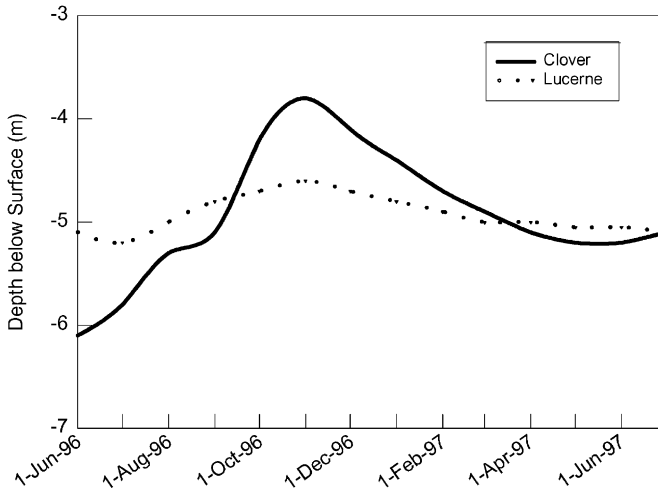


Fig. 1. Groundwater response under adjoining hill-slope treatments of clover and lucerne at the 'Ucarro' site (adapted from Hatton, 2000; personal communication).

water flow through the matrix of the subsoil within the root zone of each community. The lack of substantial lateral discharge from the hill-slopes at the 'Ucarro' site is inconsistent with conductivity values of this magnitude, but the amount of deep drainage below the root zone is consistent with the changes in groundwater height at depths of 4–6 m below the surface (Fig. 1), probably as a result of preferred pathway flow.

Discontinuities in the structure of the subsoil matrix, possibly as dykes, can intersect transmissive lateral pathways often at and above the interface of the 'A' and 'B' horizons in the soil profile. Diversion of saturated lateral flow to a vertical flux bypassing the subsoil matrix is a mechanism that contributes significantly to deep groundwater accession from the hill-slope. These conduits transfer soil water during periods of soil saturation associated with transient perched watertables. Partitioning of the excess of rainfall over plant water use into lateral and vertical outflow depends on the frequency of such conduits along the longitudinal transect of the hill-slope. In the case of the study site, diversion to vertical outflow is almost complete, reflecting high frequency of conduits along the hill-slope. By contrast, in the western sector of the 'Ucarro' farm where the interceptor banks frequently flow in winter, partitioning is akin to that of the site at BookBook in New South Wales in Australia (White et al., 2000b). At this N.S.W. site, substantial quantities of both lateral and vertical outflow were observed under perennial pasture and not unlike previous studies under annual communities in the southwest of Western Australia (Cox and McFarlane, 1995; Ward et al., 1998). At another catchment in Australia at Krawaree, New South Wales, under natural perennial grassland, a large lateral discharge was observed and only a trace of vertical flow was measured and arose from a slow diffusive re-distribution through the subsoil matrix (Aston and Dunin, 1980). Clearly, major discontinuities in subsoil structure did not occur at the Krawaree site.

The distinctive pattern of outflow observed from the hill-slope at the 'Ucarro' site can be ascribed to subsoil heterogeneity that conforms to simple one-dimensional flow for

Table 2

Total outflow by vertical drainage at 1.8 m soil depth and lateral flows (in brackets) from a 4 ha hill-slope treated with clover and lucerne at the 'Ucarro' site

| Year    | Rainfall (mm) | Clover outflow (mm) | Lucerne outflow (mm) |
|---------|---------------|---------------------|----------------------|
| 1995    | 383           | 10 (0)              | 25 (1)               |
| 1996    | 428           | 65 (1)              | 10 (2)               |
| 1997    | 323           | 25 (1)              | 2 (2)                |
| 1998    | 481           | >30 <sup>a</sup>    | 0 (0)                |
| Average | 404           | 33 (1)              | 9 (1)                |

<sup>a</sup> Clover outflow was <35 mm; adapted from Ward et al. (2001).

soil water gain up to saturation after which bypass vertical flow operates to re-charge groundwater aquifers that exist below the root zone. Independent characterization of the range of hydraulic conductivities of the heterogeneous subsoil (Hatton et al., 2002) poses a major challenge because of the issues of scale for the measurement of soil properties such as hydraulic conductivity. The studies reported in this publication come as a timely reminder that prediction of rainfall excess to generate saturated flow may be robust, but the partitioning of the excess into various forms of outflow from duplex soils is highly uncertain.

The comparison of total outflow from the hill-slopes under annual and perennial pasture indicates that reduced outflow from the perennial lucerne pasture, based on water balance measurements, has resulted from a change in vertical drainage apparently due to deeper rooting and additional water uptake (Table 2). This reduction was also expressed in the form of a diminished groundwater response (Fig. 1). The amplitude of the seasonal hydrograph in 1996 was less under lucerne (0.5 m) than under the clover hill-slope (2.4 m). This difference in amplitude of 1.9 m converts to a difference in groundwater storage of 57 mm by assuming a value of 0.03 for the specific yield of the aquifers. The agreement of this data with the water balance value of 56 mm in 1996 for the seasonal difference in vertical outflow is impressive. This consistency between independent sets of measurements for a difference in seasonal re-charge between plant communities provides evidence that drainage control by lucerne through deep uptake of soil water is achieved without major mechanical disruption to the flow pathways.

The flows in Table 2 can be used to evaluate the hydrologic consequences of incorporating lucerne into farming systems comprised of annual communities of crops and pastures. If the long-term value of deep drainage from annuals is 35 mm per year and if there is 30% lucerne on the hill-slope at any one time, the drainage from the cultivated area will be halved to 17.5 mm per year. This is comparable to a phase farming operation of 3 years in crop and 2 years in lucerne over an arable component of 80% of the hill-slope. If the proportion of lucerne on the hill-slope is 15%, the outflow will have a long-term average of 27 mm per year.

Bypass flow can also occur in the unsaturated phase of soil water gain during rain events in summer and autumn. Hydrophobic conditions contribute to instability of the wetting front causing 'fingering' and uneven soil water distribution. For one event, the 'fingering' was so pronounced that a response in deep groundwater was observed without

any detectable development of a perched watertable. This mechanism has been invoked to explain unexpected groundwater rise in response to summer thunderstorms in Western Australia. At the 'Ucarro' site it was not considered to be a significant contributor to groundwater flow, but the occurrence of this form of bypass flow emphasizes the need for caution in interpreting soil water dynamics during the period between harvest in late spring/early summer and sowing in the following autumn. Failure to measure any changes in soil water gain following rainfall events is symptomatic of a flow pattern that deviates from simple one-dimensional flow and may indicate that 'fingering' occurred. Under these circumstances, calculation of evaporative losses or deep drainage are likely to be in error.

A novel form of bypass flow with ecological significance involves the ability of the vascular system of trees at the 'Ucarro' site to re-distribute soil water both upwards in summer and downwards in winter. The upward flow termed 'hydraulic lift' (Caldwell and Richards, 1989) is now an accepted mechanism to maintain root function of shallow roots during summer. Some of the deep soil water uptake is diverted from the transpiration stream to leak through shallow root tissue for temporary storage in surface soil. The quantities of soil water involved in hydraulic lift have been difficult to determine, but its occurrence at the site is well established (Burgess et al., 1998). The measurement of water movement in the roots of the trees at the site led to the first report of reverse flow at the break of season from shallow soil depths to deep soil layers (Burgess et al., 1998). Burgess et al. (1998) termed the process hydraulic re-distribution. Retrospective analysis of soil water profiles under the tree strips have provided an estimate for the magnitude of this re-distribution (Fig. 2). Below the apparent zero flux plane at 1 m for the period of study, soil water gain increased progressively over the soil depth range

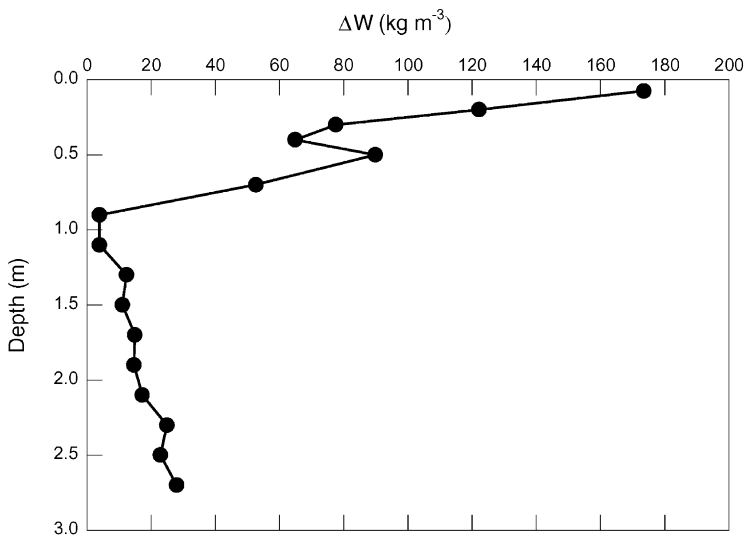


Fig. 2. Change in soil water content ( $\Delta W$ ) down the soil profile for the period 26 May to 30 June in response to rainfall of 121 mm at the 'Ucarro' site (adapted from White, 2000; personal communication).

between 1 and 3 m resulting in the accumulation of about 20 mm of water (Burgess et al., 2001). Restoration of root turgor in dry soil is a likely product of this re-distribution, ensuring that an extended profile of functional roots is available for uptake of winter rainfall that may accrue to these soil depths and minimizing the threat of accelerated deep drainage. An additional benefit of this winter re-distribution is to alleviate water logging that could impair the function of the shallow root system which is critical for seasonal transpiration.

### 3. Evapotranspiration response

“Water is the basis of all life!” (Leonardo da Vinci).

Plants use water to both grow and survive. The fate of the water for either function is loss to the atmosphere as water vapor. These contributions to atmospheric water vapor are additional to direct evaporation from soil or from intercepted water held on the canopy. For the purpose of this paper, the variation between plant communities in daily rates of total evaporation, excluding interception losses, will be explored. This is termed evapotranspiration and examination of the different losses is instructive in revealing how plant control can vary to explain some of the apparent inconsistencies related to growth that were referred to in Section 1.

Three categories of evapotranspiration response for agricultural communities and plantations, natural forest, and Mallee woodland (Dunin et al., 1999) have been observed across southern Australia (Fig. 3), and each conforms to a general form outlined by

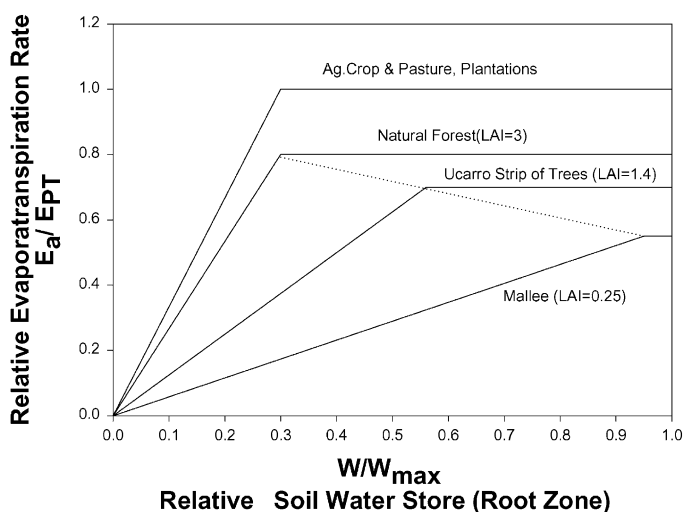


Fig. 3. Evapotranspiration characteristics derived for a range of ecosystems, each with a specified leaf area index (LAI), showing the actual daily rate of evapotranspiration ( $E_a$ ) relative to a reference rate, given by the Priestley–Taylor formula ( $E_{PT}$ ), as a function of the fractional plant available soil water in the root zone, expressed as the ratio of current soil store of plant available soil water ( $W$ ) to the total plant available soil water in the root zone ( $W_{max}$ ).

Ritchie (1981). The fourth response, designated as the ‘Ucarro strip of trees’, is drawn from observations of the present study (White et al., 2000a). The upper plateau occurs when evapotranspiration is energy limited and the values decrease below this upper limit when soil water in the root zone is limiting. The formulation of evapotranspiration in dimensionless form with relative evapotranspiration ( $E_a/E_{PT}$ ) as a function of fractional available water in the root zone is a simplified approach, but is a reasonably robust method for discriminating between plant communities and enabling the strategies involved in growth and maintenance of plant communities to be distinguished. Defining the coordinates of the ‘break point’ at the intersection of two linear segments of an individual relationship sets the critical features of the upper bound of evapotranspiration (with respect to a potential rate which, in this case, we use the Priestley–Taylor (PT) formulation,  $E_{PT}$  (Priestley and Taylor, 1972) and the soil water content that limits the rate of evapotranspiration. The relative soil water content at which soil water limits potential evapotranspiration varies with leaf area index (LAI), as reported for different stages of canopy development in a natural forest (Dunin and Aston, 1984). In a natural community with litter suppressing the rate of direct evaporation from the soil, the influence of decreasing leaf area to cause a progressive reduction in the upper boundary for the combined total rate of transpiration and direct evaporation from soil can be justified. The progressive decrease in the value of fractional soil water when the soil water supply becomes limiting is a reflection of the decreasing proportion of soil evaporation in the upper boundary as leaf area increases.

The evapotranspiration of natural vegetation differs from that of agricultural communities and plantations both in magnitude and the influence of leaf area. Plantations generally have high leaf areas (LAIs >4) which means that transpiration is dominant and stress occurs after 70% of plant available soil water is exhausted (Battaglia and Sands, 1997). Agricultural communities are similar in response. Evapotranspiration follows that of Ritchie (1981) with an upper boundary approximated by PT formulation during the growing season without any requirement to specify canopy conditions for leaf area. The insensitivity of agricultural communities to leaf area in attaining a high upper boundary is now well established for the cropping regions in Western Australia (Ward et al., 2001) and New South Wales (Dunin et al., 2001). The major reason for the maintenance of potential rates at high levels of soil water during the season of low atmospheric demand is due to complementary behavior between plant transpiration and direct evaporation from disturbed soil that is low in both organic matter and litter accumulation. During periods of high soil water content, energy not intercepted by the canopy for transpiration is dissipated at the soil surface by direct evaporation. Departures from this generalization for agricultural communities have been noted in deep sands in which soil water transport for surface evaporation can be inhibited causing the upper boundary to fall short of PT value early in the growing season. Late in the growing season, leaf area can affect the uptake rate during the phase of falling soil water reserves. However, these effects tend to be small and do not make the form of evapotranspiration response of introduced communities different from those of natural ecosystems.

The differences in evapotranspiration between introduced communities and natural ecosystems help to explain the differences in seasonal soil water patterns and growth.



Deferred water use is a feature of natural ecosystems during periods of high soil water availability. A low upper boundary for the rate of evapotranspiration and a high value for incipient stress allow soil water conservation during the season with low evaporative demand and high water use efficiency. Natural ecosystems have adapted in a way that ensures that soil water reserves are carried forward into dry periods when water use is critical for survival of perennial species. The penalty for shifting water use from times of plenty to times of shortage is a loss in primary productivity due to reduced water use efficiency which is reduced by a factor of 300–400% in dry summer months compared to wet winter months. The shift in patterns of plant water uptake accompanying the introduction of exotic communities results in the annual growth rates of these communities being an order of magnitude greater than those of natural ecosystems. The trade-off for this increased productivity is a lack of control of deep drainage in the case of annual crops and pastures due to lost summer function and a lack of deep roots to utilize the excess rainfall from the growing season that accrues below the root zone of the annuals. In the case of plantations, survival is compromised due to limited conservation of soil water during the temperate growing season.

The adaptive features for soil water conservation by natural ecosystems occurs across a range of leaf areas. For natural communities, there appears to be an association between stable leaf area and annual rainfall (Specht, 1972; Hatton et al., 1998). Features of soil water conservation are more pronounced in low annual rainfall regions such as that in the Mallee woodlands. In the present study, the belt of trees at the 'Ucarro' site had a LAI of 1.4, a value which is possibly greater than that for natural woodland due to access to groundwater (White et al., 2002). If the natural woodland has a LAI of 1.0, then the soil water conserved for uptake during the dry summer can be estimated to be 100 mm. It is because of the deferred water use that evapotranspiration of the majority of the annual rainfall occurs during summer in most years and hill-slope groundwater is maintained at a low level. This, in turn, results in low inputs of water to the riparian zone adjacent to the stream course. Maintaining a low level of the riparian watertable minimizes salt intruding into this area, whereas under conditions with extended periods of saturation in the riparian zone, salt is mobilized in the zone and salt scalds and rising stream-flow salinity occur.

Seasonal differences in plant water uptake influence plant function. Plant water uptake for transpiration during the growing season is related to biomass accumulation as crop or pasture during the season. During the remainder of the year, plant water uptake is achieved only by perennials and this is dominated by uptake of deep soil water that accumulated from excess rainfall during the preceding growing season under perennial vegetation or from several growing seasons under annual crops and pastures. The transpiration achieved, generally at low rates, is primarily for canopy cooling to overcome excessive summer heat stress and to enable persistence of the perennials through this critical period. Less of the energy is channeled for biomass accumulation, thereby lowering the water use efficiency in this season. The removal of the deep accrued soil water serves to provide a sink for ensuing deep drainage thereby mitigating threats of groundwater re-charge that may result under annual crops and pastures. An obvious means to achieve effective water use for productivity with reduced threat of re-charge is to rotate herbaceous perennials with crops.

#### 4. Integrated farming systems

“To change is to live; to change often is to be perfect” (Cardinal Newman).

Rotation of crops and pastures is an accepted means for weed and pesticide control together with enhanced nitrogen nutrition. By incorporating perennials into such rotations, greater control of deep drainage can be achieved, thereby arresting the degradation associated with groundwater rise. The papers in this publication explored different combinations of vegetation cover to evaluate how drainage and re-charge may be altered in a quest to restore the water balance to that under native vegetation.

The existing farming system at the ‘Ucarro’ farm comprises contour belts of trees covering 5% of the hill-slope with the remainder dedicated to annual crops and pastures. This combination of cover types provides the reference condition to explore the impact of expanding perennial vegetation on groundwater discharge to the riparian zone. Total outflow from annual agricultural communities is assumed to become re-charge to deep groundwater at an average rate of 35 mm per year. Groundwater discharge to the riparian zone from the hill-slope is assumed to equate to the net re-charge rate, based on the observed response of the semi-confined aquifer that returns to a former level on the annual scale (Fig. 1). Conservatively, the tree strips consume water in excess of annual rainfall to draw on groundwater at a rate of 100 mm per year (White et al., 2002). The effect of this groundwater uptake is to moderate hill-slope groundwater discharge of the transmissive aquifer by an amount determined as the product of percent projected tree area and the negative re-charge rate of 100 mm per year (i.e.  $-5$  mm per year). Scaling by the respective proportions of trees and annuals on the hill-slope, the average hill-slope groundwater discharge under existing conditions is 28 mm per year. The 0% line in Fig. 4 gives the change in groundwater discharge from this value of 28 mm with 5% tree cover

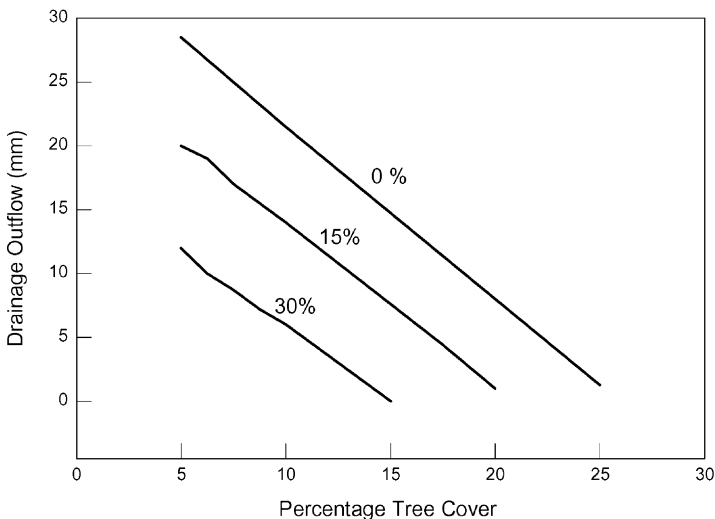


Fig. 4. Computed variation in hill-slope outflow with tree cover as contour strips for three different proportions of the land occupied by lucerne.

as the proportion of tree cover on the hill-slope increases. This line assumes that over the remaining hill-slope the agricultural component is based on annuals in rotation with no herbaceous perennials. The calculations suggest that a tree cover of 22% will be required for a leaching fraction of 5 mm per year and cover of 26% to eliminate discharge completely. Tree cover intermediate between these values is considered to provide a benign water balance for the hill-slope. Groundwater discharge in this range from the hill-slope is likely to cause the watertable in the riparian zone to retreat and enable amelioration of the salinity that has erupted along the stream course.

The 30% line in Fig. 4 represents a phase farming system with 30% of the total hill-slope occupied by the herbaceous perennial lucerne at any time. The agricultural component of the hill-slope with this proportion of lucerne is assumed to have a deep drainage outflow with an average rate of 17.5 mm per year. This might be achieved with a cropping proportion of 30% alternating with 30% lucerne, while the remaining 40% is left to annual pasture. In this system, the critical window for a benign groundwater discharge of 0–5 mm per year occurs with a tree cover between 11 and 15%. For an intermediate phase farming system with 15% lucerne and an average drainage rate of 27 mm per year, the corresponding range of tree coverage is 17–21% to ensure a benign groundwater discharge. Therefore, a range of options exists to achieve a water balance on the hill-slope that conforms to natural conditions and enables the effective control of outflow to ameliorate salinity. A phase farming system with 30% of lucerne offers the best option to maximize agricultural productivity as it minimizes the financially less-productive tree cover to 12.5%. However, it needs to be recognized that trees can contribute financially to the farming enterprise and contribute indirectly to farm productivity in several ways as outlined by the owners of ‘Ucarro’ farm (Rundle and Rundle, 2001).

The collaborative project reported in this publication has been conducted in a spirit of genuine cooperation among the participants, technical and scientific, whose generosity with unconditional data exchange has facilitated this overview. Dr. Neil C. Turner provided a cohesive framework for integrating the separate studies that enabled the trade-offs between enhanced water control and productivity to be evaluated. A significant catalyst for the integration has been the additional funds to CCMAR provided by Dr. John Radcliffe, deputy chief executive of CSIRO, who complemented the industry funding already acknowledged in papers of this special issue. Finally, we are grateful to the Rundle family who happily tolerated the frequent visits to their property ‘Ucarro’ so that the farm management strategies adopted to counter water logging and salinity could be scientifically characterized and evaluated.

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